Questions by topic: The hydrosphere: hydrological cycle & unsustainable exploitation - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the volume of water in some reservoirs of the hydrological cycle and the annual transfer rates through them.

Table 1

Reservoir	**Volume / x 10^{15} dm^3**	**Annual transfer (inflow = outflow) / x 10^{15} dm^3 yr^{-1}**	**Percentage of global water / %**
Oceans	1 340 000	425	97
Ice caps and glaciers	24 000	3	1.7
Groundwater	23 000	12	1.7
Lakes and rivers	180	40	0.013
Soil moisture	17	80	0.001
Atmosphere	13	480	0.001

0 1 . 1 The residence time of water in a reservoir is given by: residence time = volume in reservoir / annual transfer rate. Use Table 1 to calculate the residence time of water in the atmosphere. Give your answer to the nearest whole day.

[3 marks]

Answer _____ days

0 1 . 2 Calculate the residence time of water in the oceans.

[1 mark]

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Answer [ ] years

0 1 . 3 Explain why the atmosphere is an important reservoir even though it holds only 0.001% of global water.

[2 marks]

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0 1 . 4 Explain why human changes to a small reservoir such as soil moisture are more obvious than changes to the oceans.

[2 marks]

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0 1 . 5 Explain why the hydrological cycle is described as being in dynamic equilibrium.

[2 marks]

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0 2 . 1 Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation .....

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Urban development .....

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Agriculture .....

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0 2 . 2 Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

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0 2 . 3 Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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0 3 The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

Table 1

Year	1960	1980	2000	2015
Lake area / km^2	68 000	52 000	24 000	7 000
Salinity / g dm^{-3}	10	18	45	over 100
Commercial fish catch / thousand tonnes	40	10	0	0

0 3 . 1 Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

[2 marks]

Answer ____ %

0 3 . 2 Explain why the salinity of the lake increased as the area fell.

[2 marks]

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0 3 . 3 Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

[4 marks]

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0 3 . 4 Suggest why disagreements over the rivers continue between the countries of the region.

[2 marks]

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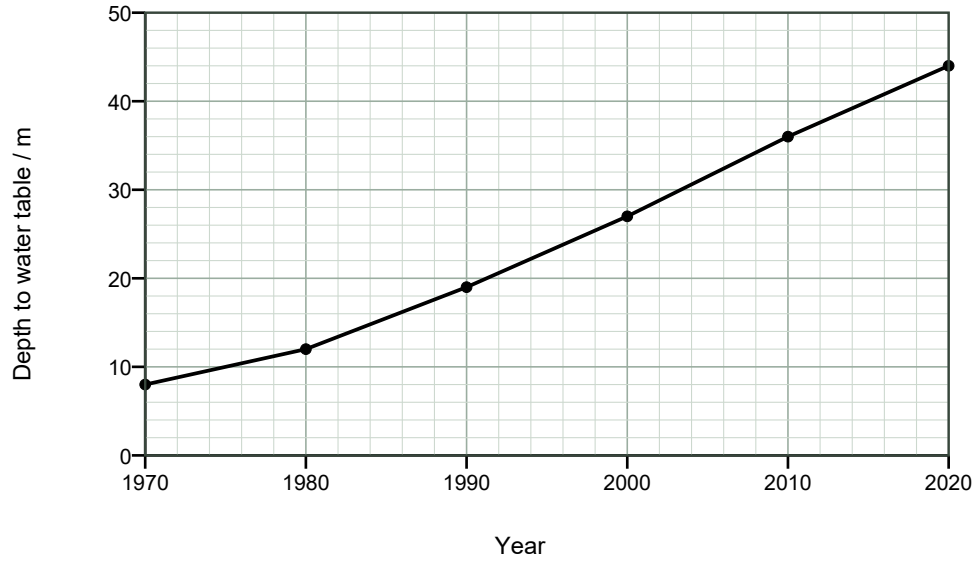
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0 4 Figure 1 shows the depth of the water table below the ground surface in an aquifer beneath a coastal farming region between 1970 and 2020.

Figure 1



0 4 . 1 Calculate the mean rate at which the water table fell between 1970 and 2020.

[1 mark]

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Answer [ ] m/yr

0 4 . 2 Explain why the water table is falling.

[2 marks]

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0 4 . 3 Explain the features a rock must have to form an aquifer.

[3 marks]

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0 4 . 4 Describe **four** consequences of over-exploiting this coastal aquifer.

[4 marks]

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0 5 . 1 Explain why abstraction from an aquifer containing 'fossil' (palaeo) water, such as the High Plains / Ogallala Aquifer in the USA, cannot be sustainable.

[3 marks]

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0 5 . 2 Describe how the GRACE satellites are used to monitor aquifer depletion and give **one** advantage over monitoring boreholes.

[3 marks]

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0 5 . 3 Describe **three** measures being introduced on the North China Plain to reduce aquifer depletion.

[3 marks]

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0 5 . 4 Suggest why large-scale afforestation on the North China Plain may reduce aquifer recharge even though trees reduce soil erosion.

[1 mark]

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0 6 A dam and reservoir are to be built on a river to supply water and regulate flow.

0 6 . 1 Explain **three** effects the reservoir may have on wildlife in and around the river.

[6 marks]

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0 6 . 2 Explain why decomposition of flooded vegetation in a new reservoir contributes to climate change.

[2 marks]

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0 6 . 3 Explain how a river-regulation reservoir reduces both flooding and water shortages downstream.

[2 marks]

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0 7 An aquifer beneath an agricultural plain holds 3.6×10^{11} m^3 of water. It is recharged at 1.2×10^9 m^3 per year, and 4.8×10^9 m^3 per year is abstracted from it.

0 7 . 1 Residence time = volume of the reservoir / rate of flow through it. Calculate the residence time of the water in this aquifer, using the recharge rate.

[2 marks]

Answer _____ years

0 7 . 2 Explain the properties a rock must have to form a useful aquifer.

[3 marks]

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0 7 . 3 Explain why the water table beneath the plain is falling.

[2 marks]

0 7 . 4 Describe **four** consequences of continuing to over-abstract this aquifer.

[4 marks]

0 7 . 5 Explain why an aquifer holding 'fossil water' cannot be exploited sustainably, and describe **two** measures that would reduce abstraction.

[4 marks]

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END OF QUESTIONS

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